

Software Requirements & Dependencies

Environment Configuration & Dependency Specification

Manuscript: "Graphene-Assisted Electrostatic Interface Engineering for Radiation-Tolerant MR-DWELL Dosimeters"

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This document specifies the software dependencies, environment setup, and system requirements necessary to execute the computational modeling, numerical simulation, and machine-learning classification scripts in the Supplementary Material repository.

1. Recommended requirements.txt File

Save the following exact package specifications into a file named requirements.txt in the root directory of the software package:

```
# Core numerical and scientific computing libraries
numpy>=1.21.0
scipy>=1.7.0      # Added for integration and numerical solvers

# Visualization and plotting engine
matplotlib>=3.5.0

# Machine learning and multivariate classification
scikit-learn>=1.0.0
```

2. Detailed Package Justification & Functional Mapping

Package	Min Version	Functional Role & Script Usage
numpy CORE	1.21.0	Array vectorization, matrix algebra, physical constants handling, and numerical evaluations in analytic_model.py .
scipy RECOMMENDED	1.7.0	Special functions, transmission integral evaluations, numerical integration, and optimization algorithms across plot_fig1.py and plot_fig2.py .
matplotlib CORE	3.5.0	Publication-quality figure generation (Figures S1, S2, S3), vector graphic exports (PDF/EPS), and color mapping.
scikit-learn CORE	1.0.0	Dimensionality reduction (PCA/LDA) and radiation classification pipeline in classification_model.py .

3. Environment Setup & Installation Guide

Option A: Quick Installation using Standard Pip

```
# Create virtual environment (optional but recommended)
python -m venv env
source env/bin/activate # On Windows: env\Scripts\activate

# Install specified dependencies
pip install -r requirements.txt
```

Option B: Conda Environment Setup

```
# Create a clean Conda environment
conda create -n mrdwell_env python=3.9 numpy scipy matplotlib scikit-learn -y
conda activate mrdwell_env
```

Verification & Environment Check

To verify that all dependencies are correctly installed and that the analytical model executes as expected, run:

```
python analytic_model.py
```

This executes the built-in self-tests and validates physical constants and capacitance screening factors.